In [18]:
```
from ucimlrepo import fetch_ucirepo
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
iris = fetch_ucirepo(id=53)
```

In [6]:
```
iris
```

Out[6]:
```
{'data': {'ids': None,
  'features':      sepal length  sepal width  petal length  petal width
  0             5.1          3.5           1.4          0.2
  1             4.9          3.0           1.4          0.2
  2             4.7          3.2           1.3          0.2
  3             4.6          3.1           1.5          0.2
  4             5.0          3.6           1.4          0.2
  ..            ...          ...           ...          ...
  145           6.7          3.0           5.2          2.3
  146           6.3          2.5           5.0          1.9
  147           6.5          3.0           5.2          2.0
  148           6.2          3.4           5.4          2.3
  149           5.9          3.0           5.1          1.8

  [150 rows x 4 columns],
  'targets':               class
  0       Iris-setosa
  1       Iris-setosa
  2       Iris-setosa
  3       Iris-setosa
  4       Iris-setosa
  ..              ...
  145  Iris-virginica
  146  Iris-virginica
  147  Iris-virginica
  148  Iris-virginica
  149  Iris-virginica

  [150 rows x 1 columns],
  'original':      sepal length  sepal width  petal length  petal width
class
  0             5.1          3.5           1.4          0.2     Iris-setosa
  1             4.9          3.0           1.4          0.2     Iris-setosa
  2             4.7          3.2           1.3          0.2     Iris-setosa
  3             4.6          3.1           1.5          0.2     Iris-setosa
  4             5.0          3.6           1.4          0.2     Iris-setosa
  ..            ...          ...           ...          ...             ...
  145           6.7          3.0           5.2          2.3  Iris-virginica
  146           6.3          2.5           5.0          1.9  Iris-virginica
  147           6.5          3.0           5.2          2.0  Iris-virginica
  148           6.2          3.4           5.4          2.3  Iris-virginica
  149           5.9          3.0           5.1          1.8  Iris-virginica

  [150 rows x 5 columns],
  'headers': Index(['sepal length', 'sepal width', 'petal length', 'petal width',
'class'], dtype='object')},
 'metadata': {'uci_id': 53,
  'name': 'Iris',
  'repository_url': 'https://archive.ics.uci.edu/dataset/53/iris',
  'data_url': 'https://archive.ics.uci.edu/static/public/53/data.csv',
  'abstract': 'A small classic dataset from Fisher, 1936. One of the earliest know
n datasets used for evaluating classification methods.\n',
  'area': 'Biology',
  'tasks': ['Classification'],
  'characteristics': ['Tabular'],
  'num_instances': 150,
```

```
  'num_features': 4,
  'feature_types': ['Real'],
  'demographics': [],
  'target_col': ['class'],
  'index_col': None,
  'has_missing_values': 'no',
  'missing_values_symbol': None,
  'year_of_dataset_creation': 1936,
  'last_updated': 'Tue Sep 12 2023',
  'dataset_doi': '10.24432/C56C76',
  'creators': ['R. A. Fisher'],
  'intro_paper': {'ID': 191,
   'type': 'NATIVE',
   'title': 'The Iris data set: In search of the source of virginica',
   'authors': 'A. Unwin, K. Kleinman',
   'venue': 'Significance, 2021',
   'year': 2021,
   'journal': 'Significance, 2021',
   'DOI': '1740-9713.01589',
   'URL': 'https://www.semanticscholar.org/paper/4599862ea877863669a6a8e63a3c707a7
87d5d7e',
   'sha': None,
   'corpus': None,
   'arxiv': None,
   'mag': None,
   'acl': None,
   'pmid': None,
   'pmcid': None},
  'additional_info': {'summary': 'This is one of the earliest datasets used in the
literature on classification methods and widely used in statistics and machine lea
rning.  The data set contains 3 classes of 50 instances each, where each class ref
ers to a type of iris plant.  One class is linearly separable from the other 2; th
e latter are not linearly separable from each other.\n\nPredicted attribute: class
of iris plant.\n\nThis is an exceedingly simple domain.\n\nThis data differs from
the data presented in Fishers article (identified by Steve Chadwick,  spchadwick@e
speedaz.net ).  The 35th sample should be: 4.9,3.1,1.5,0.2,"Iris-setosa" where the
error is in the fourth feature. The 38th sample: 4.9,3.6,1.4,0.1,"Iris-setosa" whe
re the errors are in the second and third features.  ',
   'purpose': 'N/A',
   'funded_by': None,
   'instances_represent': 'Each instance is a plant',
   'recommended_data_splits': None,
   'sensitive_data': None,
   'preprocessing_description': None,
   'variable_info': None,
   'citation': None}},
 'variables':            name     role         type demographic  \
 0  sepal length  Feature   Continuous        None
 1   sepal width  Feature   Continuous        None
 2  petal length  Feature   Continuous        None
 3   petal width  Feature   Continuous        None
 4         class   Target  Categorical        None

                                          description units missing_values
 0                                               None    cm             no
 1                                               None    cm             no
```

```
2                                                None    cm             no
3                                                None    cm             no
4  class of iris plant: Iris Setosa, Iris Versico...  None             no  }
```

In [8]:
```
df = iris.data.features
```

In [9]:
```
df.isnull().sum()
```

Out[9]:
```
sepal length    0
sepal width     0
petal length    0
petal width     0
dtype: int64
```

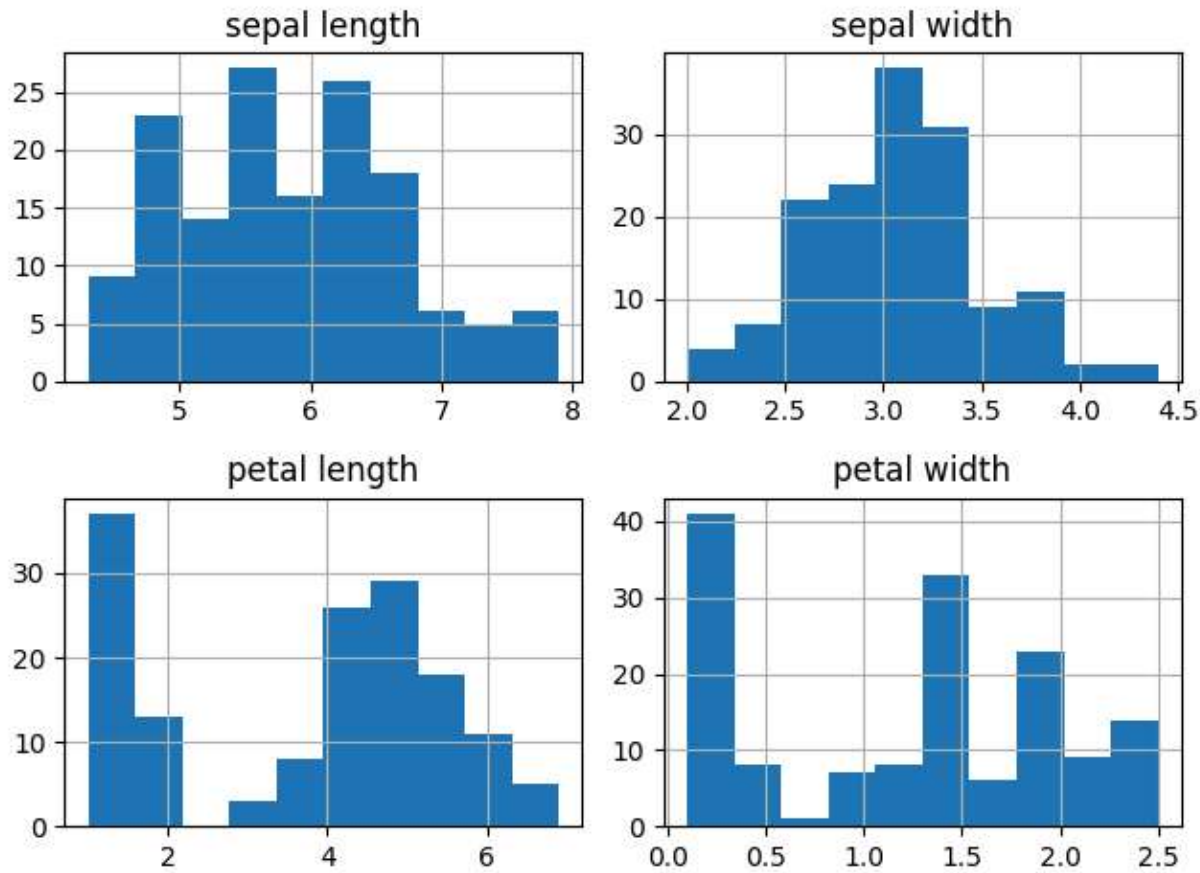
In [10]:
```
df['species'] = iris.data.targets
```

In [11]:
```
df.describe()
```

Out[11]:

	sepal length	sepal width	petal length	petal width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [16]:
```
df.hist()
plt.tight_layout()
```

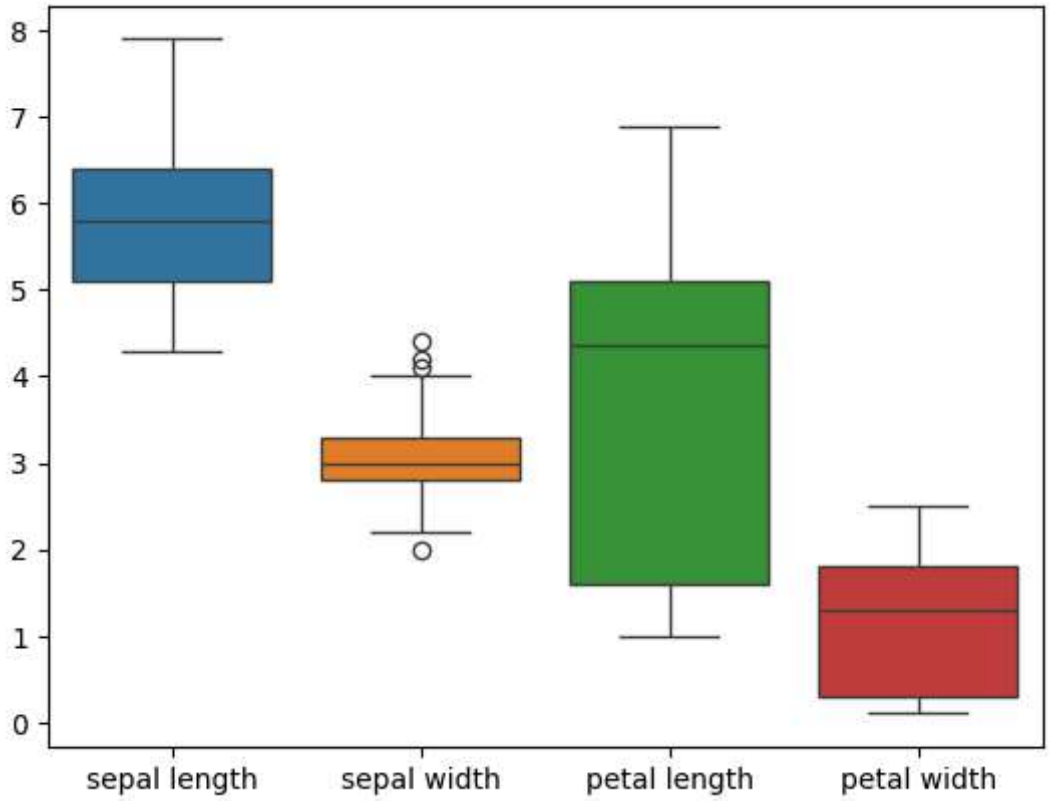


In [17]:
```
df['sepal length'].mode()
```

Out[17]:
```
0    5.0
Name: sepal length, dtype: float64
```

In [21]:
```
sns.boxplot(data=df.iloc[:, 0:4])
```

Out[21]:
```
<Axes: >
```


8
7
6
5
4
3
2
1
0
sepal length
sepal width
petal length
petal width

In [ ]: